N2 — The Apex Determinant: A Closed Form via Helmert–Fourier Factorization

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Abstract

For each integer $N \geq 2$ let $U_N \in U(N-1)$ be the unitary matrix that intertwines the normalized Helmert basis and the nontrivial additive characters of $\mathbb{Z}/N$ on the mean-zero hyperplane of $\mathbb{C}^N$. We prove the closed form

$$\det U_N = i^{-\binom{N-1}{2}},$$

exhibiting the determinant as a fourth root of unity that depends only on N modulo 8. The proof realizes U_N as the nontrivial diagonal block of the product HF^{T} of the full $N \times N$ Helmert matrix H and the symmetric Fourier matrix F, which reduces $\det U_N$ to $\det H \cdot \det F$. We evaluate $\det H = (-1)^{N-1}$ in closed form by a cofactor recursion, and recall Schur's evaluation of the Fourier determinant. As arithmetic consequences we determine the exact field of the determinant: $\det U_N$ is a fourth root of unity for every N, equals ± 1 when $N \equiv 1, 2 \pmod 4$ and $\pm i$ when $N \equiv 0, 3 \pmod 4$, and lies in the cyclotomic field $\mathbb{Q}(\zeta_N)$ if and only if $N \not\equiv 3 \pmod 4$. The period-8 dependence on N refines Schur's period-4 formula by exactly the Helmert reflection sign.

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1. Introduction

The matrix studied here arises as the change of basis, on the mean-zero hyperplane of $\mathbb{C}^N$, between two natural orthonormal frames: the real Helmert frame, built from the nested contrasts $(1, \dots, 1, -k, 0, \dots, 0)$, and the complex Fourier frame of nontrivial additive characters of $\mathbb{Z}/N$. The resulting unitary $U_N \in U(N-1)$ has entries in the compositum $K_N \cdot \mathbb{Q}(\zeta_N)$ of the multiquadratic field $K_N = \mathbb{Q}(\sqrt{p} : p \le N)$ and the cyclotomic field $\mathbb{Q}(\zeta_N)$; its entries individually carry both quadratic-radical and cyclotomic content. The companion paper [N1] establishes the basic structure of the family $\{U_N\}$ and computes $\det U_4 = i$ by direct expansion.

This paper determines $\det U_N$ for all N in closed form.

Theorem 1.1. For every integer $N \ge 2$,

$$\det U_N = i^{-\binom{N-1}{2}} = i^{-(N-1)(N-2)/2}.$$

In particular $\det U_N \in \{1, -1, i, -i\}$ for every N, and the value is periodic with period 8 in N:

$N \bmod 8$	0	1	2	3	4	5	6	7
$\det U_N$	$-i$	1	1	$-i$	i	-1	-1	i

The proof is a factorization. Embedding U_N into the full $N \times N$ Helmert and Fourier matrices realizes it as a diagonal block of their product (Proposition 3.1), so that $\det U_N = \det H \cdot \det F$. The Helmert factor is an elementary sign, $\det H = (-1)^{N-1}$ (Lemma 4.1), and the Fourier factor is the classical Schur determinant of the discrete Fourier transform (Lemma 5.1). Assembling the two and reducing the exponent modulo 4 yields Theorem 1.1 (Section 6).

The closed form has clean arithmetic corollaries. The determinant is always a fourth root of unity (Corollary 7.1); it is real (± 1) exactly for $N \equiv 1, 2 \pmod 4$ and imaginary ($\pm i$) exactly for $N \equiv 0, 3 \pmod 4$ (Corollary 7.2); and although every entry of U_N involves quadratic radicals, the determinant lies in the cyclotomic field $\mathbb{Q}(\zeta_N)$ precisely when $N \not\equiv 3 \pmod 4$ (Corollary 7.4). The last statement isolates the single residue class on which the multiquadratic content of the entries fails to cancel within $\mathbb{Q}(\zeta_N)$, and we identify the obstruction with the conductor of an explicit quadratic field (Proposition 7.5).

Throughout, $\omega = e^{2\pi i/N}$ and $\zeta_m = e^{2\pi i/m}$. We write $\binom{n}{2} = n(n-1)/2$.

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2. The apex unitary

We fix the conventions used for the rest of the paper. They agree with the closed form of [N1, T9] and with the explicit matrices computed there.

2.1 Two orthonormal frames of the hyperplane

Let $H_N := \{v \in \mathbb{C}^N : \sum_{j=0}^{N-1} v_j = 0\}$, with the standard Hermitian inner product $\langle u, v\rangle = \sum_j u_j \overline{v_j}$.

The **Helmert vectors** h_k $(k = 1, \dots, N-1)$ consist of k leading ones, then the entry $-k$, then $N-1-k$ trailing zeros:

$$h_k := (\,1, \dots, 1,\, -k,\, 0, \dots, 0\,) \in \mathbb{Z}^N, \qquad k = 1, \dots, N-1,$$

which are real, mutually orthogonal, mean-zero, with $\|h_k\|^2 = k(k+1)$; the normalized vectors $e_k := h_k/\sqrt{k(k+1)}$ form an orthonormal basis of H_N.

The **character vectors** are $\chi_c := (\omega^{cj})_{j=0}^{N-1}$ for $c = 1, \dots, N-1$; they are mean-zero and orthogonal, with $\|\chi_c\|^2 = N$, and the normalized $\tilde{\chi}_c := \chi_c/\sqrt{N}$ form a second orthonormal basis of H_N.

2.2 Definition and entry formula

Definition 2.1. The *apex unitary at level N* is the change-of-basis matrix $U_N \in \mathbb{C}^{(N-1)\times(N-1)}$ with entries

$$(U_N)_{kc} := \langle \tilde{\chi}_c, e_k\rangle = \frac{1}{\sqrt{Nk(k+1)}} \left(\sum_{j=0}^{k-1} \omega^{cj} - k\,\omega^{ck} \right), \qquad k, c \in \{1, \dots, N-1\}.$$

Because e_k is real, $\langle \tilde{\chi}_c, e_k\rangle = \frac{1}{\sqrt{N}} \sum_j \omega^{cj}\,(e_k)_j = R_k \sum_j h_k(j)\,\omega^{cj}$, where

$$R_k := \frac{1}{\sqrt{Nk(k+1)}}.$$

Summing the geometric series gives the equivalent closed form

$$\boxed{(U_N)_{kc} = R_k \cdot \frac{1 - (k+1)\,\omega^{kc} + k\,\omega^{(k+1)c}}{1 - \omega^c}} \qquad (2.1)$$

used throughout. Since U_N is the matrix of an orthonormal change of basis, $U_N \in U(N-1)$.

Remark 2.2 (orientation of the pairing). The determinant is sensitive to the conjugation convention. Definition 2.1 pairs the *character* $\tilde{\chi}_c$ against the (real) Helmert vector e_k, i.e. it is conjugate-linear in the character; equivalently it pairs h_k bilinearly with $\chi_c(j) = \omega^{cj}$. This is the convention reproduced by the closed form (2.1) and by the explicit level-4 matrix of [N1]. The opposite pairing $\langle e_k, \tilde{\chi}_c\rangle = \overline{(U_N)_{kc}}$ yields the complex-conjugate matrix $\overline{U_N}$, which is also unitary but has $\det \overline{U_N} = \overline{\det U_N}$; the two conventions differ by complex conjugation and hence give determinants $i^{\mp\binom{N-1}{2}}$.

2.3 The radical–cyclotomic split

Each entry factors as $(U_N)_{kc} = R_k\, C_{kc}$ with $R_k \in K_N$ depending only on k, and

$$C_{kc} = \frac{1 - (k+1)\omega^{kc} + k\omega^{(k+1)c}}{1 - \omega^c} \in \mathbb{Z}[\omega] \subset \mathbb{Q}(\zeta_N).$$

Thus the quadratic radicals enter only through the row scalars R_k, and the cyclotomic content only through the integer matrix $C = (C_{kc})$. This split governs the field of the determinant in Section 7.

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3. The block factorization

We pass to the full $N \times N$ picture, including the mean direction and the trivial character.

Let H be the **full normalized Helmert matrix**: its rows, indexed $0, 1, \dots, N-1$, are

$$\text{row } 0 = \tfrac{1}{\sqrt{N}}(1, \dots, 1), \qquad \text{row } k = e_k \ (1 \le k \le N-1).$$

H is real orthogonal. Let F be the **symmetric unitary Fourier matrix**

$$F_{cj} = \frac{\omega^{cj}}{\sqrt{N}}, \qquad c, j = 0, \dots, N-1,$$

so $F = F^{\mathsf{T}}$ and F is unitary.

Proposition 3.1 (block factorization). The product $M := HF^{\mathsf{T}}$ has the block form

$$M = \begin{pmatrix} 1 & 0 \\ 0 & U_N \end{pmatrix},$$

where the upper-left entry is the $(0,0)$ scalar, the off-diagonal blocks are zero, and the trailing $(N-1) \times (N-1)$ block is U_N in the convention of Definition 2.1. Consequently

$$\det U_N = \det M = \det H \cdot \det F.$$

Proof. For $0 \le r, c \le N-1$,

$$M_{rc} = \sum_{j=0}^{N-1} H_{rj}\, F_{cj} = \sum_j H_{rj}\, \frac{\omega^{cj}}{\sqrt{N}}.$$

Row 0. With $H_{0j} = 1/\sqrt{N}$, $M_{0c} = \frac{1}{N}\sum_j \omega^{cj} = \delta_{c0}$, since $\sum_j \omega^{cj} = N$ for $c \equiv 0$ and 0 otherwise.

Column 0. With $F_{0j} = 1/\sqrt{N}$, $M_{r0} = \frac{1}{\sqrt{N}}\sum_j H_{rj}$. The row sums of H are $\sum_j H_{0j} = \sqrt{N}$ and, for $k \ge 1$, $\sum_j H_{kj} = \frac{1}{\sqrt{k(k+1)}}\sum_j h_k(j) = 0$. Hence $M_{00} = 1$ and $M_{k0} = 0$ for $k \ge 1$.

Trailing block. For $k, c \ge 1$,

$$M_{kc} = \sum_j \frac{h_k(j)}{\sqrt{k(k+1)}} \cdot \frac{\omega^{cj}}{\sqrt{N}} = R_k \sum_j h_k(j)\, \omega^{cj} = (U_N)_{kc}$$

by Definition 2.1. Therefore $M = 1 \oplus U_N$, and $\det M = \det U_N$. Finally $\det M = \det H \det F^{\mathsf{T}} = \det H \det F$. $\square$

This is the precise sense in which U_N is "the discrete Fourier transform restricted to nontrivial modes, read in the Helmert frame": the mean mode and the trivial character split off as the 1×1 corner, and what remains is exactly U_N.

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4. The Helmert determinant

Lemma 4.1. $\det H = (-1)^{N-1}$.

Proof. Write $H = D\,\tilde{H}$, where $D = \mathrm{diag}(1/\sqrt{N},\, 1/\sqrt{1 \cdot 2},\, \ldots,\, 1/\sqrt{(N-1)N})$ and $\tilde{H}$ is the integer matrix with row $0 = (1, \ldots, 1)$ and row $k = h_k$ for $1 \le k \le N-1$. Then $\det H = \det D \cdot \det \tilde{H}$.

The scalar. Using $\prod_{k=1}^{N-1} k(k+1) = (N-1)!\,N!$ and $N \cdot (N-1)! = N!$,

$$\det D = \Big(N \prod_{k=1}^{N-1} k(k+1)\Big)^{-1/2} = \big((N!)^2\big)^{-1/2} = \frac{1}{N!}.$$

The integer determinant. We claim

$$\det \tilde{H}_N = (-1)^{N-1} N!. \qquad (4.1)$$

We prove (4.1) by establishing the exact recursion $\det \tilde{H}_N = -N \det \tilde{H}_{N-1}$ for $N \ge 2$, with base $\det \tilde{H}_1 = 1$.

Expand $\det \tilde{H}_N$ along the last column (column $N-1$). Its entries are: 1 in row 0 (the all-ones row); 0 in rows $1, \ldots, N-2$ (the vector h_k is supported on columns $0, \ldots, k$ with $k \le N-2$); and $-(N-1)$ in row $N-1$ (the pivot of h_{N-1}). Hence the Laplace expansion has only two terms,

$$\det \tilde{H}_N = 1 \cdot (-1)^{0+(N-1)}\, M_{0,N-1} \;+\; \big(-(N-1)\big) \cdot (-1)^{(N-1)+(N-1)}\, M_{N-1,N-1}, \qquad (4.2)$$

where $M_{r,c}$ denotes the (r,c) minor and $(-1)^{2(N-1)} = 1$.

The minor $M_{N-1,N-1}$ deletes the last row and column, leaving rows $0, \ldots, N-2$ over columns $0, \ldots, N-2$. Row 0 is the all-ones row of length $N-1$, and each h_k with $1 \le k \le N-2$ is unchanged (its support lies in columns $0, \ldots, k$). This is exactly $\tilde{H}_{N-1}$:

$$M_{N-1,N-1} = \det \tilde{H}_{N-1}.$$

The minor $M_{0,N-1}$ deletes row 0 and the last column, leaving rows $1, \ldots, N-1$ over columns $0, \ldots, N-2$. The rows $h_1, \ldots, h_{N-2}$ are unchanged; the row $h_{N-1} = (1, \ldots, 1, -(N-1))$ loses its last entry, becoming the all-ones row of length $N-1$. Thus this block has rows $(h_1, \ldots, h_{N-2}, \mathbf{1})$, while $\tilde{H}_{N-1}$ has rows $(\mathbf{1}, h_1, \ldots, h_{N-2})$. The two differ by moving the top row past the $N-2$ rows below it, i.e. by $N-2$ transpositions, so

$$M_{0,N-1} = (-1)^{N-2} \det \tilde{H}_{N-1}.$$

Substituting into (4.2) and using $(-1)^{(N-1)+(N-2)} = (-1)^{2N-3} = -1$,

$$\det \tilde{H}_N = (-1)^{N-1}(-1)^{N-2} \det \tilde{H}_{N-1} - (N-1) \det \tilde{H}_{N-1} = -\det \tilde{H}_{N-1} - (N-1) \det \tilde{H}_{N-1} = -N \det \tilde{H}_{N-1}.$$

With $\det \tilde{H}_1 = 1$ this telescopes to $\det \tilde{H}_N = \prod_{m=2}^{N}(-m) = (-1)^{N-1}N!$, proving (4.1).

Therefore $\det H = \det D \cdot \det \tilde{H}_N = \frac{1}{N!} \cdot (-1)^{N-1}N! = (-1)^{N-1}$. $\square$

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5. The Fourier determinant

The determinant of the symmetric Fourier matrix is classical (Schur, 1921). We record the evaluation in the unreduced form that is convenient for assembly.

Lemma 5.1. $\det F = i^{\,E_N}$, where $E_N = \binom{N}{2} + (N-1)^2 = \dfrac{N(N-1)}{2} + (N-1)^2$.

Proof. Let $V = \sqrt{N}\,F$, so $V_{cj} = \omega^{cj} = (\omega^c)^j$ is the Vandermonde matrix in the nodes $\omega^0, \omega^1, \dots, \omega^{N-1}$. Hence

$$\det V = \prod_{0 \le a < b \le N-1} (\omega^b - \omega^a).$$

Using $e^{i\theta} - 1 = 2i \sin(\theta/2)\, e^{i\theta/2}$ with $\theta = 2\pi(b-a)/N$,

$$\omega^b - \omega^a = \omega^a(\omega^{b-a} - 1) = 2i\, \sin\frac{\pi(b-a)}{N}\, e^{\,i\pi(a+b)/N}.$$

There are $\binom{N}{2}$ factors, so

$$\det V = (2i)^{\binom{N}{2}} \Big(\prod_{a<b} \sin \tfrac{\pi(b-a)}{N} \Big) \exp\Big(\frac{i\pi}{N} \sum_{a<b} (a+b) \Big).$$

Each index appears in $N-1$ unordered pairs, whence $\sum_{a<b}(a+b) = (N-1)\sum_{m=0}^{N-1} m = \frac{N(N-1)^2}{2}$, and the exponential equals $e^{\,i\pi(N-1)^2/2} = i^{(N-1)^2}$. Separating the modulus from the phase,

$$\det V = \rho_N \cdot i^{\binom{N}{2}} \cdot i^{(N-1)^2}, \qquad \rho_N := 2^{\binom{N}{2}} \prod_{a<b} \sin \tfrac{\pi(b-a)}{N} \;>\; 0.$$

The factor ρ_N is a positive real number; since $V = \sqrt{N}\,F$ with F unitary, its value is $\rho_N = |\det V| = (\sqrt{N})^N = N^{N/2}$. Therefore

$$\det V = N^{N/2}\, i^{\binom{N}{2}+(N-1)^2}, \qquad \det F = \frac{\det V}{N^{N/2}} = i^{\binom{N}{2}+(N-1)^2} = i^{\,E_N}. \qquad \square$$

(Each $\sin \frac{\pi m}{N}$ with $1 \le m \le N-1$ is positive, so the modulus is fixed by unitarity and no sign ambiguity enters; only the explicit phase i^{E_N} remains.)

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6. Proof of Theorem 1.1

Combining Proposition 3.1 with Lemmas 4.1 and 5.1,

$$\det U_N = \det H \cdot \det F = (-1)^{N-1}\, i^{\,E_N} = i^{\,2(N-1)+E_N},$$

since $(-1)^{N-1} = i^{2(N-1)}$. It remains to reduce the exponent modulo 4 to the target $-\binom{N-1}{2}$. Compute the difference exactly as a polynomial in N:

$$\Big(2(N-1)+E_N\Big) - \Big(-\tbinom{N-1}{2}\Big) = 2(N-1) + \frac{N(N-1)}{2} + (N-1)^2 + \frac{(N-1)(N-2)}{2}.$$

The two half-integer terms combine to $\frac{(N-1)\big(N+(N-2)\big)}{2} = (N-1)^2$, so the difference equals

$$2(N-1) + 2(N-1)^2 = 2(N-1)\big(1+(N-1)\big) = 2N(N-1).$$

Because $N(N-1)$ is a product of consecutive integers it is even, hence $2N(N-1) \equiv 0 \pmod 4$. As i has order 4,

$$\det U_N = i^{\,2(N-1)+E_N} = i^{-\binom{N-1}{2}}. \qquad \blacksquare$$

As a check, at $N = 4$: $\det H = (-1)^3 = -1$, $E_4 = 6 + 9 = 15$, $\det F = i^{15} = -i$, so $\det U_4 = (-1)(-i) = i$, agreeing with the direct computation of [N1].

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7. Arithmetic consequences

We record the consequences of Theorem 1.1 for the value, the periodicity, and the field of the determinant. Write $D_N := \binom{N-1}{2} = \frac{(N-1)(N-2)}{2}$, so $\det U_N = i^{-D_N}$.

Corollary 7.1 (torsion). For every $N \geq 2$, $\det U_N$ is a fourth root of unity; in particular U_N has determinant of finite order dividing 4, and $\det U_N \in \{1, -1, i, -i\}$.

Proof. Immediate from $\det U_N = i^{-D_N}$ with $D_N \in \mathbb{Z}$. $\square$

Corollary 7.2 (real vs. imaginary). $\det U_N \in \{1, -1\}$ if $N \equiv 1, 2 \pmod 4$, and $\det U_N \in \{i, -i\}$ if $N \equiv 0, 3 \pmod 4$. Equivalently, $\det U_N$ is imaginary iff D_N is odd.

Proof. $\det U_N = i^{-D_N}$ is real iff D_N is even and imaginary iff D_N is odd. Reducing $D_N = \frac{(N-1)(N-2)}{2}$ modulo 2 by cases on N mod 4: for $N \equiv 1$, $N-1 \equiv 0$ (4) so D_N is even; for $N \equiv 2$, $N-2 \equiv 0$ (4) so D_N is even; for $N \equiv 0$, $(N-1)$ is odd and $(N-2)/2$ is odd, so D_N is odd; for $N \equiv 3$, both $(N-1)/2$ and $N-2$ are odd, so D_N is odd. $\square$

Corollary 7.3 (periodicity). The map $N \mapsto \det U_N$ has minimal period 8 in N, as tabulated in Section 1.

Proof. $\det U_N$ depends only on $D_N \bmod 4$, and $D_N = \binom{N-1}{2}$ is a quadratic in N; a quadratic modulo 4 has period dividing 8, and the tabulated values show all eight residues are realized, so the period is exactly 8. $\square$

We now turn to the field generated by the determinant. Recall that the group of roots of unity contained in $\mathbb{Q}(\zeta_N)$ is μ_N if N is even and μ_{2N} if N is odd; in particular $i \in \mathbb{Q}(\zeta_N)$ if and only if $4 \mid N$.

Corollary 7.4 (cyclotomic field of the determinant). $\det U_N \in \mathbb{Q}(\zeta_N)$ if and only if $N \not\equiv 3 \pmod 4$. In the exceptional case $N \equiv 3 \pmod 4$ one has $\det U_N = \pm i \in \mathbb{Q}(\zeta_4) \setminus \mathbb{Q}(\zeta_N)$, and $\det U_N \in \mathbb{Q}(\zeta_{4N})$.

Proof. By Corollary 7.2, $\det U_N \in \{\pm 1\} \subset \mathbb{Q} \subset \mathbb{Q}(\zeta_N)$ when $N \equiv 1, 2 \pmod 4$. When $N \equiv 0, 3 \pmod 4$ we have $\det U_N = \pm i$, which lies in $\mathbb{Q}(\zeta_N)$ iff $i \in \mathbb{Q}(\zeta_N)$ iff $4 \mid N$. Among $N \equiv 0, 3 \pmod 4$

this holds for $N \equiv 0$ and fails for $N \equiv 3$. Combining the cases, $\det U_N \notin \mathbb{Q}(\zeta_N)$ exactly when $N \equiv 3 \pmod 4$. There $\det U_N = \pm i \in \mathbb{Q}(\zeta_4)$, and $\mathbb{Q}(\zeta_4), \mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}(\zeta_{4N})$. $\square$

Corollary 7.4 is striking in view of the entry structure of U_N: every entry carries multiquadratic radicals through the scalars R_k, yet the determinant collapses to a fourth root of unity that is cyclotomically clean except on a single residue class. The next proposition locates the obstruction precisely, in terms of the squarefree part of an explicit integer.

Proposition 7.5 (the radical obstruction). Let

$$P_N := \prod_{k=1}^{N-1} Nk(k+1) = N^{\,N-1}\,(N-1)!\,N!,$$

and let $\mathrm{sf}(P_N)$ denote its squarefree part. Then

$$\det U_N \in \mathbb{Q}(\zeta_N) \iff \sqrt{\mathrm{sf}(P_N)} \in \mathbb{Q}(\zeta_N) \iff \mathrm{cond}(\mathbb{Q}(\sqrt{\mathrm{sf}(P_N)})) \mid N,$$

where $\mathrm{cond}(\mathbb{Q}(\sqrt{m})) = m$ if $m \equiv 1 \pmod 4$ and $4m$ otherwise (and 1 if $m = 1$). In particular, combined with Corollary 7.4, the conductor divides N precisely when $N \not\equiv 3 \pmod 4$.

Proof. By the split of Section 2.3, $\det U_N = \big(\prod_{k=1}^{N-1} R_k\big) \det C$ with $\det C \in \mathbb{Q}(\zeta_N)$, since C has entries in $\mathbb{Z}[\omega]$. The scalar is

$$\prod_{k=1}^{N-1} R_k = \prod_{k=1}^{N-1} \frac{1}{\sqrt{Nk(k+1)}} = \frac{1}{\sqrt{P_N}} = \frac{1}{q\sqrt{\mathrm{sf}(P_N)}}$$

for a positive rational q (write $P_N = q^2\,\mathrm{sf}(P_N)$). As U_N is unitary, $\det C = \det U_N \cdot q\sqrt{\mathrm{sf}(P_N)} \neq 0$. Hence $\det U_N = \det C/(q\sqrt{\mathrm{sf}(P_N)})$ lies in $\mathbb{Q}(\zeta_N)$ iff $\sqrt{\mathrm{sf}(P_N)} \in \mathbb{Q}(\zeta_N)$, because $\det C \in \mathbb{Q}(\zeta_N)$ and the field is closed under nonzero division. The quadratic field $\mathbb{Q}(\sqrt{m})$ is contained in $\mathbb{Q}(\zeta_N)$ iff its conductor divides N, by the conductor–discriminant correspondence for quadratic subfields of cyclotomic fields; this gives the stated conductor criterion. The final clause is Corollary 7.4. $\square$

Proposition 7.5 is the precise form of the "radicals cancel" phenomenon: the multiquadratic radicals appearing throughout the entries of U_N contribute to the determinant only through the single radical $\sqrt{\mathrm{sf}(P_N)}$, and this radical is cyclotomic—so the determinant is cyclotomically clean—exactly when $N \not\equiv 3 \pmod 4$. The criterion is genuinely about the *combined* radical $\sqrt{\mathrm{sf}(P_N)}$ rather than its prime factors individually: distinct non-cyclotomic prime radicals may combine into a cyclotomic one, as at $N = 21$, where $\mathrm{sf}(P_{21}) = 21 \equiv 1 \pmod 4$ has conductor $21 \mid 21$ even though neither $\sqrt{3}$ nor $\sqrt{7}$ lies in $\mathbb{Q}(\zeta_{21})$.

Remark 7.6 (relation to Schur's formula). Schur's determinant $\det F = i^{E_N}$ depends on N only modulo 4 (the exponent $E_N = \binom{N}{2} + (N-1)^2$ is quadratic in N, but its parity pattern realizes only four values). The apex determinant $\det U_N = i^{-D_N}$ depends on N modulo 8 (Corollary 7.3). The refinement from period 4 to period 8 is supplied entirely by the Helmert reflection sign $(-1)^{N-1} = i^{2(N-1)}$, whose exponent $2(N-1)$ is the odd-order term that doubles the period. Thus the additional arithmetic content of U_N over the bare Fourier transform is exactly one factor of the orientation of the real Helmert frame.

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8. Summary

The determinant of the apex unitary is the fourth root of unity $\det U_N = i^{-\binom{N-1}{2}}$, established by the factorization $\det U_N = \det H \cdot \det F$ into an elementary Helmert sign and Schur's Fourier determinant. The value is periodic of period 8 in N; it is a square root of unity for $N \equiv 1, 2 \pmod 4$ and a primitive fourth root for $N \equiv 0, 3 \pmod 4$; and it is cyclotomically clean—lying in $\mathbb{Q}(\zeta_N)$—if and only if $N \not\equiv 3 \pmod 4$, the obstruction in the remaining class being the single radical $\sqrt{\mathrm{sf}(N^{N-1}(N-1)!\,N!)}$.

The trace and the spectrum of U_N, whose analysis is governed by the conductor decomposition of $\mathbb{Z}/N$ rather than by a single determinantal sign, are taken up in subsequent papers of the series.

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